

Right Ventricular Metrics as End Points in Clinical Trials

A Review

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IMPORTANCE Clinical trials in cardiovascular medicine aim to deliver high-quality evidence with greater efficiency, including smaller sample sizes and shorter timelines. The selection of sensitive and reliable end points is central to this goal. Right ventricular (RV) structure and function play a critical role in many cardiovascular conditions; however, RV-specific parameters remain underused and inconsistently applied in interventional clinical trials. In this article, the current and emerging role of RV-focused imaging and hemodynamic parameters as end points in clinical trials are explored and a comprehensive framework for their development, validation, and broader implementation is proposed.

OBSERVATIONS Across multiple disease domains, RV-focused parameters derived from echocardiography, cardiac magnetic resonance imaging, computed tomography, and right-sided heart catheterization demonstrate potential value in assessing the therapeutic effects of new medicines or devices across a wide range of conditions including heart failure, cardiomyopathies, valve diseases, pulmonary hypertension, and congenital heart conditions. However, their use is inconsistent, and feasibility challenges are common, particularly in multicenter clinical trial settings. Key limitations include variability in imaging acquisition, incomplete data capture, lack of standardized protocols, and insufficient validation linking short-term RV changes to long-term clinical outcomes. Emerging tools such as artificial intelligence may improve reproducibility and efficiency, while collaborative, multidisciplinary efforts are essential for advancing end-point development.

CONCLUSIONS AND RELEVANCE RV-specific end points hold substantial promise for enhancing the sensitivity and mechanistic insight of clinical trials but remain inadequately integrated into current practice. Optimal end-point selection requires balancing clinical relevance, analytical validity, and operational feasibility, with consideration of disease-specific pathophysiology and trial design. Standardization of acquisition, validation of clinically meaningful changes, and centralized data analysis are important next steps. Broader adoption of robust RV metrics supported by technological innovation and cross-sector collaboration has the potential to improve trial efficiency, accelerate therapeutic development, and ultimately enhance patient outcomes.

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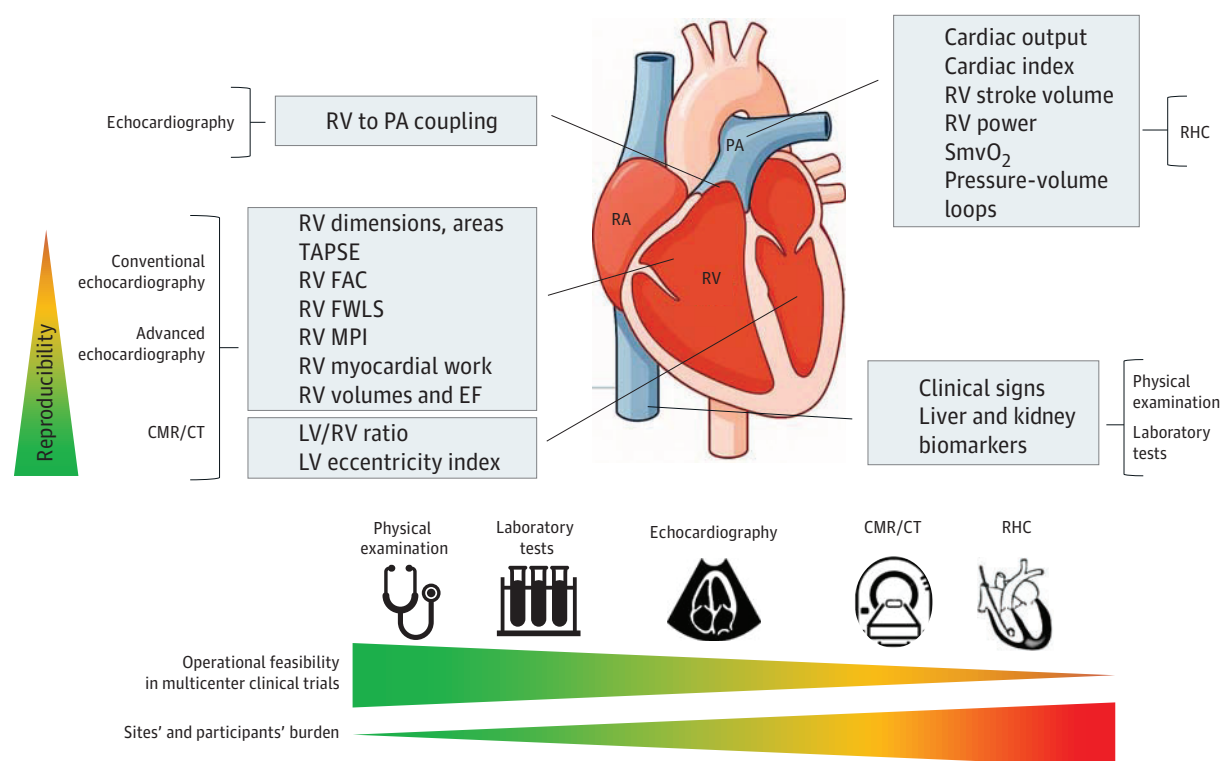
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The contemporary fundamentals of clinical trials have elevated the rigor and quality of biomedical research to standards that ensure an exceptional quality of evidence to enhance everyday clinical practice.¹ Still, there are ongoing efforts to further optimize the value of clinical studies given the current landscape of accelerated cardiovascular drug and device discovery among major pharmaceutical and biotechnology companies, driven by aggressive innovation pipelines, breakthrough therapies, and novel advanced drug delivery modalities. The ability to run clinical trials with smaller sample sizes, rapid enrollment, and shorter treatment durations but unaltered or increased efficacy signal detection ability is critical.² The appropriate selection of the study end points is of paramount importance in this process.

Well-selected, disease-specific imaging or laboratory biomarkers play an important role in early-phase clinical trials to investigate positive mechanistic effects of interventions, evaluate safety and efficacy signals, and mitigate the risk of required major investment, intrinsic to phase 3 trials.³

In patients with congenital heart disease (CHD), right-sided valvular heart disease, pulmonary arterial hypertension (PAH) or pulmonary hypertension due to left heart disease, the importance of right ventricular (RV) morphology and function is crucial, requiring the inclusion of RV-oriented clinical measures, especially in early trials. RV-specific end points may be also highly relevant in left heart diseases, where studies have shown that RV dysfunction provides independent and incremental prognostic information beyond conventional left ventricular (LV) metrics.^{4,5} Accordingly, there is a clear

Figure. Diagram Depicting the Imaging, Hemodynamic, and Laboratory Biomarkers of Right Ventricular Remodeling and Function, With Considerations for Clinical Trial Use



CMR indicates cardiac magnetic resonance imaging; CT, computed tomography; EF, ejection fraction; FAC, fractional area change; FWLS, free-wall longitudinal strain; LV, left ventricle/ventricular; MPI, myocardial performance index; PA, pulmonary artery; RA, right atrium; RHC, right-sided heart catheterization; RV, right ventricle/ventricular; SmvO₂, mixed venous oxygen saturation; TAPSE, tricuspid annular plane systolic excursion.

and pressing need to develop, validate, and systematically incorporate measures of RV function and RV-pulmonary artery coupling into the design and analysis of clinical trials.

Current Use of RV-Specific End Points in Interventional Clinical Trials

A wide range of imaging- and hemodynamic-based parameters have been used in clinical practice to assess RV properties, and some have been tested in clinical trials. Noninvasive imaging modalities (echocardiography, cardiac magnetic resonance [CMR], and cardiac computed tomography [CT]) are most frequently used in clinical trials for assessment of RV remodeling and function,⁶ and right-sided heart catheterization (RHC) is commonly used to provide supplementary hemodynamic information.⁷ Although RV-specific circulating biomarkers are lacking, laboratory indicators of end-organ damage related to RV dysfunction or venous congestion may also be useful measures. The Venous Excess Ultrasound (VEXUS) score has been validated for assessing venous congestion and end-organ damage and correlates with elevated central venous pressure and RV-pulmonary artery (RVPA) coupling.^{8,9}

The most important markers reflecting RV remodeling and function that have been used in clinical trials are detailed in the Figure and eTable 1 in the Supplement.¹⁰⁻²⁶

To evaluate the current implementation of RV parameters as end points in interventional clinical trials, we conducted a structured literature review of clinical trials that incorporated any metric of the RV across a broad range of cardiopulmonary diseases in the last 15 years. The detailed description of methods, inclusion criteria, and search strategy is provided in the eMethods in the Supplement.¹⁰⁻²⁶ Main findings are summarized in eTables 2 to 5 in the Supplement.¹⁰⁻²⁶

RV Parameters in Heart Failure Clinical Trials

Despite the increasing recognition of the impact of RV remodeling and impaired function on outcomes in HF, its incorporation into clinical trials remains inconsistent, and RV parameters have been assessed in a small fraction of the trials conducted in the last 15 years (eTable 2 in the Supplement).¹⁰⁻²⁶ This illustrates both the potential value and practical challenges of incorporating RV imaging end points into clinical trials.

The Effects of Dapagliflozin on Cardiac Deformation and Clinical Outcomes in Heart Failure With Reduced and Mildly Reduced Ejection Fraction (DAPA ECHO) trial²⁷ demonstrated that advanced echocardiographic measures such as RV free-wall longitudinal strain (FWLS) and global longitudinal strain (GLS) are feasible as well as modifiable in response to treatment with dapagliflozin in heart failure with reduced ejection fraction (HFrEF) and HF with mildly reduced EF (HFmrEF). Significant improvements in RV strain were

observed at 6 months, highlighting the sensitivity of deformation imaging. This was, however, a single-center clinical trial allowing for standardized acquisition and rigorous quality control. The study population had abnormal values of RV strain at baseline, allowing room for measurable change. Although strain is a promising technique in highly standardized settings, pitfalls of FWLS include inter-vendor variability,²⁸ test-retest variability (which may be as high as $2.2\% \pm 1.9\%$),²⁹ and the imaging acquisition view (higher values from an RV-focused vs standard 4-chamber view).³⁰

By contrast, 2 larger multicenter studies in HF with preserved EF (HFpEF), such as Soluble Guanylate Cyclase Stimulator in Heart Failure Patients With Preserved EF (SOCRATES-PRESERVED) (vericiguat)³¹ and the Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist (TOPCAT) echocardiographic substudy (spironolactone),³² reported neutral findings. RV indices such as tricuspid annular systolic excursion (TAPSE) and RV fractional area change (FAC) showed no meaningful differences between treatment arm and placebo. Importantly, measurability was suboptimal, as less than half of the randomized participants had analyzable RV FAC at follow-up. Short treatment duration (12 weeks) in the SOCRATES-PRESERVED trial and the relatively preserved RV function at baseline in both studies posed challenges to detecting therapeutic effects.

The most recent echocardiography substudy was of the Effect of Semaglutide 2.4 mg Once Weekly on Function and Symptoms in Subjects With Obesity-Related HFpEF (STEP-HFpEF) program, which enrolled 491 participants with HFpEF and obesity, evaluating RV morphology and function as exploratory efficacy end points.³³ Treatment with semaglutide for 52 weeks resulted in statistically significant reductions in RV areas compared with placebo. However, no measurable effects on RV functional parameters, such as TAPSE, FAC, and S' (ie, tissue Doppler imaging S' velocity), were observed. Importantly, RV areas and FAC could be quantified at both baseline and follow-up in only 26% of participants, which is possibly related to challenges with the echocardiographic acoustic window in a population of patients with obesity.³³

Small trials (N <50) have evaluated RVPA coupling during exercise. In the Evaluation of the Cardiac and Metabolic Effects of Dapagliflozin in Heart Failure With Preserved Ejection Fraction (CAMEO-DAPA) trial, dapagliflozin led to significant improvements in RVPA coupling assessed by S' velocity indexed to PA pressure during exercise in patients with HFpEF, effects that were not apparent at rest.³⁴

Overall, TAPSE, RV areas, and FAC, S', and RV strain are the echocardiographic parameters most commonly used in HF clinical trials in recent years. RV strain seems to be a sensitive marker of therapeutic response in the population with HFrEF in single-center study design settings. However, significant challenges exist regarding the feasibility of using these parameters in multicenter phase 2 and 3 trials. Important limitations of these measurements include standardization of imaging acquisition, which contributes to test-retest measurement variability.³⁰

Changes in RV function were also investigated in trials evaluating mechanical circulatory support. RV failure is a frequent complication of LV assist device (LVAD) implantation, prolonging intensive care unit admissions, and occasionally necessitating temporary RV assist device (RVAD) support, associated with poor outcomes.³⁵ LVAD trials have used composite end points to

define RV failure, which often incorporate the requirement for RVAD or prolonged inotropic or pulmonary vasodilator support, as well as hemodynamic markers of severe RV functional deterioration. Importantly, the decision to initiate RVAD support is frequently challenging and influenced by limitations in RV assessment and ongoing debate regarding the most appropriate metrics to define clinically meaningful RV failure.

RV Parameters in Clinical Trials of Cardiomyopathies

RV involvement is increasingly recognized in cardiomyopathies; however, its systematic evaluation in clinical trials is limited to recent studies in hypertrophic cardiomyopathy (HCM) and transthyretin amyloid cardiomyopathy (ATTR-CM) (eTable 2 in the [Supplement](#)).¹⁰⁻²⁶

In obstructive HCM, the A Randomized, Double-blind, Placebo-Controlled Study to Evaluate Mavacamten in Adults With Symptomatic Obstructive Hypertrophic Cardiomyopathy Who Are Eligible for Septal Reduction Therapy (VALOR-HCM) trial³⁶ included RV echocardiographic parameters such as TAPSE, RV GLS, and FWLS in patients treated with mavacamten. Right-sided heart end points also included RV volumes, although echocardiographic technical details on the methods used to assess RV volumes were not provided. RV strain parameters demonstrated very high feasibility. Absence of RV strain worsening provided important safety information in response to chronic treatment with this cardiac myosin inhibitor. The A Phase 3, Multicenter, Randomized, Double-Blind, Placebo-Controlled Trial to Evaluate the Efficacy and Safety of CK-3773274 in Adults With Symptomatic Hypertrophic Cardiomyopathy and Left Ventricular Outflow Tract Obstruction (SEQUOIA-HCM) trial³⁷ included TAPSE and S' as safety markers during aficamten therapy. Both indices declined modestly, although values remained within normal ranges.

In ATTR-CM, the A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of Vutrisiran in Patients With Transthyretin Amyloidosis With Cardiomyopathy (HELIOS-B) trial^{38,39} incorporated a broader panel of RV end points, including wall thickness, areas, FAC, and S' measured by echocardiography. Over both 18- and 30-month follow-up, vutrisiran attenuated the decline in RV systolic function, with significant preservation of RV S' and borderline effects on FAC.

Overall, recent examples of RV echocardiographic parameters used in clinical trials of HCM and ATTR amyloidosis (RV free-wall thickness, areas, TAPSE, S', FAC, RV GLS, and FWLS) demonstrated their feasibility and potential value in the evaluation of safety and efficacy of novel therapies in these populations.

RV Parameters in Clinical Trials of Transcatheter Valve Interventions

The RV plays a central role in the prognosis of valvular heart disease, including after intervention.^{40,41} The role of the RV in the maintenance of circulatory homeostasis has become increasingly clear in the era of transcatheter interventions, as we repair and replace valves in the beating heart in patients who are frail and multimorbid. Despite this, assessment of RV parameters is far from a standard approach in trials. Although seldom reported in transcatheter aortic valve replacement or mitral transcatheter edge-to-edge-repair trials, metrics for RV function have begun to appear in trials for transcatheter tricuspid valve intervention; however, RV size or function has not been used as an end point in these trials.

The landmark Clinical Trial to Evaluate Cardiovascular Outcomes in Patients Treated With the Tricuspid Valve Repair System (TRILUMINATE) Pivotal trial investigated the effects of tricuspid transcatheter edge-to-edge-repair.⁴² Although this trial did not report specific RV function parameters at baseline and 1-year follow-up, a 2-year follow-up study demonstrated significant changes from baseline in TAPSE, RV end-diastolic diameter, FAC and GLS.⁴³ Importantly, changes in RV function are not linear, with an initial decline in functional parameters, followed by gradual recovery over time.⁴⁴ The Edwards EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device (TRISCEND II) trial of a transcatheter tricuspid valve replacement system reported baseline and 1-year data for TAPSE, RV end-diastolic diameter, indexed RV end-diastolic volume (assessed by 3-dimensional [3D] echocardiography), RV outflow tract stroke volume, TAPSE/pulmonary artery systolic pressure, and FAC,⁴⁵ demonstrating improvement in RV remodeling and function; however, data were available for only a portion of the study population, eg, in as few as 10.7% of cases for indexed RV end-diastolic volume, reflecting challenges with the feasibility of 3D echocardiography when used in multicenter trial settings (eTable 3 in the Supplement).¹⁰⁻²⁶

In East Asia, prospective single-arm studies have evaluated 2 tricuspid interventions: the K-clip (Shanghai Huihe Healthcare Technology Co Ltd), a transcatheter annuloplasty device,^{46,47} and the LuX valve (Jenscare Scientific Co Ltd),⁴⁸ a transcatheter tricuspid valve replacement. Across studies, RV size decreased (septolateral annular dimension and RV end-diastolic diameter), but indices of RV systolic function (FAC and TAPSE) did not show significant improvement at 1 year. A potential limitation was the absence of echocardiography-derived RV EF and RV strain analyses, which might have been feasible at least in a single-center trial design. Without these, regional and nonlongitudinal mechanics may be overlooked, sensitivity to early functional change is reduced, and measurements remain more dependent on angle and loading conditions.

Reported changes in RV function and remodeling in response to TV interventions have been supported by the results of multiple registries,^{49,50} demonstrating improvements in RV dimensions and function (eTable 6 in the Supplement).¹⁰⁻²⁶

In summary, available data consistently demonstrate the remodeling effect of tricuspid valve intervention, underscoring the importance of incorporating RV indices as formal end points in future trials. The current challenges lie in identifying which parameters are most clinically meaningful in the context of RV volume overload (where early postinterventional changes in RV indices may reflect acute alterations in loading conditions rather than true changes in intrinsic RV contractility), as well as measures that remain feasible in multicenter clinical trial settings, while providing maximal prognostic and mechanistic insight.

In contrast, RV function has not been assessed systematically as end points in clinical trials of transcatheter aortic and mitral interventions, despite clear evidence from observational and registry studies demonstrating its prognostic value. Studies reporting RV parameters in the context of transcatheter aortic and mitral valve intervention are listed in eTables 7 and 8 in the Supplement, respectively.¹⁰⁻²⁶

RV Parameters in Clinical Trials in PAH

RV pressure overload and failure are the most important determinants of symptomatic status and prognosis in PAH.⁵¹ RHC-based end points are frequently used in clinical trials (eTable 4 in the Supplement).¹⁰⁻²⁶ Cardiac output and/or cardiac index, attributable to improved RV function in response to interventions in PAH, were commonly reported RV end points considered to track with primary/key secondary end points in the corresponding phase 3 clinical trials. Being fundamental parameters of RHC protocols, they display generally high feasibility in clinical trial settings. On the other hand, the typically unspecified method of determination (eg, thermodilution or [in]direct Fick method) and the lack of core laboratory analysis may affect the robustness of the measurement.

In contrast, cardiac imaging end points are scarce in the PAH clinical trial domain, with only a few trials investigating echocardiography and CMR RV parameters. Thus, a phase 3 trial with imatinib reported on echocardiography measures of RV, demonstrating improvements in RV function (RV S', myocardial performance index, RV outflow tract velocity time integral).⁵² Trials of sotatercept demonstrated significant change in FAC and TAPSE/pulmonary artery systolic pressure—an important surrogate for RVPA coupling—but not TAPSE alone), as assessed by echocardiography, and reduction in RV volumes, as assessed by CMR.^{53,54} A phase 4 open-label single-arm study specifically monitored the impact of a medical treatment strategy on RV remodeling and function in PAH, and used CMR-derived RV stroke volume as a coprimary end point (along with RHC-derived change in pulmonary vascular resistance), showing improvements in both parameters.⁵⁵ Notably, a recently proposed concept integrates favorable changes in right-sided heart hemodynamics and imaging findings to provide a clinically meaningful end point for PAH trials; however, this approach requires further validation.⁵⁶

RV Parameters in Clinical Trials in Pulmonary Embolism

Pulmonary CT angiography is the mainstay diagnostic method for the confirmation of pulmonary embolism,⁵⁷ although providing information on cardiac dimensions, and the LV/RV ratio is the most commonly used RV end point in clinical trials (eTable 4 in the Supplement),¹⁰⁻²⁶ with established prognostic value in this population.⁵⁸ An echocardiography-derived LV/RV ratio and quantitative (eg, TAPSE) and semiquantitative (RV dysfunction by visual assessment, Pulmonary Embolism International Thrombolysis [PEITHO] criteria of unrecovered RV)⁵⁹ parameters have also been reported. Despite the less robust reproducibility and the evidently inferior nature of semiquantitative RV functional measures, the predominance of this approach persists even in the recent and ongoing trials in the field.⁶⁰

RV Parameters in Clinical Trials in Congenital Heart Disease

The critical role of the RV in determining disease course is well established in CHD.⁶¹ Several relatively small clinical trials have used RV morphology and function as primary and secondary efficacy end points, such as RVEF, RV volumes, and mass (eTable 5 in the Supplement).^{10-26,62-66}

CMR has been the predominant imaging modality, with CT used in cases where CMR was contraindicated.^{65,66} The yield of these CMR- or CT-derived parameters was high, with end points being measurable typically in greater than 90% of the study population. A clinical trial of valsartan demonstrated a significant reduction in RV vol-

umes and mass in adult patients with a systemic RV, without significant improvement in RVEF.⁶⁶ Dapagliflozin treatment also resulted in statistically significant improvement in RV FAC and FWLS in failing systemic RV, as assessed by echocardiography.⁶⁷ The Safety and Efficacy of Dapagliflozin in Adult Patients With a Systemic Right Ventricle (DAPA-SERVE) trial uniquely included 3D echocardiography-derived RVEF.⁶⁷ Across other trials, pharmacological interventions, such as tadalafil in patients with a systemic RV^{65,66} and losartan and ramipril in patients with repaired tetralogy of Fallot,^{62,63} failed to demonstrate significant treatment effects on RV remodeling and function. Similarly, surgical approaches, including pulmonary valve replacement combined with RV remodeling surgery, did not achieve a difference in RVEF.⁶⁴ Notably, robust randomized data incorporating RV-specific end points in broader cardiac surgical populations (beyond LVAD and congenital heart disease) remain limited—an important gap that warrants explicit acknowledgment and further investigation.

Framework for Development, Validation, and Wider Use of RV-Specific End Points

Main Principles of RV-Specific End-Point Selection

The development of RV-specific end points for clinical trials requires a careful balance between clinical validity (ie, an end point should be specific to the underlying pathophysiology, clinically relevant, modifiable by intervention, and linked to outcomes or changes in function and symptoms), analytical validity (an end point should be reproducible and sufficiently sensitive to detect small change in RV structure and function during relatively short treatment duration), and operational feasibility, frequently in multicenter clinical trial settings.⁶⁸⁻⁷⁰ Each domain highlights distinct challenges that must be addressed when selecting parameters.

From an analytical standpoint, RV parameters are inherently more variable than LV indices due to the RV's complex geometry, load dependence, and frequent limitations in image quality. Volumetric measures derived from 3D echocardiography or CMR are among the most reproducible indices of RV morphology and function. Although they provide a more comprehensive assessment than conventional echocardiographic metrics, their acquisition and analysis are not yet fully standardized.^{69,71} Longitudinal strain parameters, although sensitive to early changes in RV function, can still be affected by interobserver variability.²⁸ It remains unclear whether clinical trials should prioritize test-retest reproducibility or sensitivity in detecting early changes in response to intervention. The optimal strategy may vary depending on the sample size, trial design, and expected effect size. In this context, rigorous analytical standardization and quality assurance become particularly critical. The validity and reproducibility of right-sided heart end points in clinical trials critically depend on centralized core laboratory-based analysis, harmonized analytic standards, rigorous and continuous quality control, and structured training with real-time feedback mechanisms, given the inherently greater variability and more limited prior use of RV measurements compared with LV indices.

From a clinical validity perspective, evidence (frequently lacking) that short-term changes in RV parameters predict long-term changes in outcomes, hence clinical benefit, is needed. Furthermore, thresholds for minimal clinically meaningful change remain

incompletely defined, underscoring the need for rigorous validation studies to establish which metrics best serve as trial end points in different contexts. An additional and important physiological domain—RV diastolic dysfunction—remains largely neglected in this context, despite its relevance to disease progression and outcomes; notably, the recently published 2025 American Society of Echocardiography guidelines on right-sided heart evaluation explicitly reemphasize this concept, particularly in pulmonary hypertension and CHD.⁷²

Operational feasibility presents additional constraints. Some of the most reproducible measures, such as CMR-derived volumes, may not be practical for large-scale trials due to their high costs and limited accessibility. Conversely, echocardiographic indices, although more widely available, are more variable and dependent on the acoustic window and operator expertise.

Finally, population-specific considerations are essential.⁷² The prognostic and mechanistic relevance of RV parameters differ across disease states (eg, pulmonary hypertension, HFpEF, CHD). A parameter that is highly informative in one setting may be less meaningful in another.⁷¹ This suggests that RV end points should not be selected as one size fits all, but rather tailored to the population and intervention under investigation. Future research should therefore focus on building evidence for disease- and context-specific RV end points, while also exploring composite or multidimensional algorithms.⁶⁹

Role of Artificial Intelligence

Artificial intelligence (AI) and machine learning hold significant promise for advancing the use of RV end points in clinical trials. Automated segmentation and quantification algorithms can substantially reduce observer dependency, thereby improving reproducibility and precision—an especially valuable gain in RV assessment.^{73,74} By streamlining labor-intensive steps in data processing, AI has the potential to lower both the time and cost burdens of large-scale analyses, which are often limiting factors in multicenter trials. Importantly, these tools may be leveraged not only for end point evaluation but also during screening, facilitating more efficient trial enrolment.

Beyond automation, AI may also serve as a catalyst for end point discovery, identifying novel imaging biomarkers or patterns of RV function that predict outcomes but elude conventional analysis. In the near term, hybrid workflows that integrate human expertise with AI-driven analysis—particularly within core laboratories—are likely to represent the most pragmatic approach, combining the efficiency of automation with the interpretive oversight of experienced clinicians.

Cross-Functional Collaboration to Drive Innovation in RV Evaluation in Clinical Trials

The successful development and implementation of RV-specific imaging and hemodynamic end points in clinical trials will benefit from close collaboration within a diverse group of stakeholders, including clinical trialists, cardiologists, pulmonologists, imaging and catheterization specialists, engineers, software developers, and partners from the pharmaceutical and device industries.

Academic-industry partnerships can further accelerate the validation of novel end points and facilitate the regulatory approval process by providing the necessary scientific and technical

evidence. Shared training platforms and knowledge transfer initiatives will enhance consistency across sites and disciplines, while promoting broader adoption of best practices.

Such collaborative frameworks are critical for advancing RV imaging from exploratory research tools to robust, validated end points that can meaningfully inform therapeutic development and patient care.

Conclusions

Although RV dysfunction plays a pivotal role in prognosis, its assessment in clinical trials of new medicines or devices remains relatively limited, with considerable variability in the use and feasibility of imaging and hemodynamic RV end points. Trials in heart failure, cardiomyopathies, PAH, valvular heart disease, and CHD have shown that RV met-

rics can provide valuable insights into therapeutic responses. From a clinical validity perspective, end-point selection should reflect RV pathophysiology within each clinical context, reflect the primary therapeutic mechanism (eg, afterload reduction or reverse remodeling), and have established links to clinical outcomes. From an analytical validity point of view, priority should be given to RV measures with demonstrated reproducibility and standardized acquisition and analysis protocols. Finally, operational feasibility, especially in multicenter clinical trial settings, is another important consideration essential to minimize missing data and need for imputation.

Future efforts should be focused on standardization of RV assessment, development and validation of reliable, reproducible end points that are both clinically meaningful, sensitive to detect change, and feasible across diverse patient populations, to provide deep mechanistic insight and ultimately enhance the evaluation of novel therapies and improve patient care.

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REFERENCES

1. DeMets DL, Califf RM. A historical perspective on clinical trials innovation and leadership: where have the academics gone? *JAMA*. 2011;305(7):713-714. doi:[10.1001/jama.2011.175](https://doi.org/10.1001/jama.2011.175)

2. Zannad F, Pitt B. The future of clinical trials. *Circulation*. 2024;149(23):1783-1785. doi:[10.1161/CIRCULATIONAHA.123.066982](https://doi.org/10.1161/CIRCULATIONAHA.123.066982)

3. Hahn RT, Lawlor MK, Davidson CJ, et al. Tricuspid valve academic research consortium definitions for tricuspid regurgitation and trial end points. *Eur Heart J*. 2023;44(43):4508-4532. doi:[10.1093/eurheartj/ehad653](https://doi.org/10.1093/eurheartj/ehad653)

4. Kitano T, Bartoš F, Nabeshima Y, Sayour AA, Kovács A, Takeuchi M. Impact of cardiovascular magnetic resonance-derived right ventricular ejection fraction on adverse outcomes: a robust bayesian model-averaged meta-analysis. *J Cardiovasc Magn Reson*. 2024;26(2):101118. doi:[10.1016/j.jcmr.2024.101118](https://doi.org/10.1016/j.jcmr.2024.101118)

5. Surkova E, Muraru D, Genovese D, Aruta P, Palermo C, Badano LP. Relative prognostic importance of left and right ventricular ejection fraction in patients with cardiac diseases. *J Am Soc Echocardiogr*. 2019;32(11):1407-1415.e3. doi:[10.1016/j.echo.2019.06.009](https://doi.org/10.1016/j.echo.2019.06.009)

6. Gillam LD, Leipsic J, Weissman NJ. Use of imaging endpoints in clinical trials. *JACC Cardiovasc Imaging*. 2017;10(3):296-303. doi:[10.1016/j.jcmg.2016.12.003](https://doi.org/10.1016/j.jcmg.2016.12.003)

7. Rosenkranz S, Preston IR. Right heart catheterization: best practice and pitfalls in pulmonary hypertension. *Eur Respir Rev*. 2015;24(138):642-652. doi:[10.1183/16000617.0062-2015](https://doi.org/10.1183/16000617.0062-2015)

8. Longino A, Martin K, Leyba K, et al. Prospective evaluation of venous excess ultrasound for estimation of venous congestion. *Chest*. 2024;165(3):590-600. doi:[10.1016/j.chest.2023.09.029](https://doi.org/10.1016/j.chest.2023.09.029)

9. Rola P, Souligny WB, Haycock K, Spiegel R, Denault A. VEXUS and right ventricular to pulmonary artery uncoupling. *Crit Care*. 2025;30(1):56. doi:[10.1186/s13054-025-05806-9](https://doi.org/10.1186/s13054-025-05806-9)

10. Shah AM, Claggett B, Sweitzer NK, et al. Prognostic importance of changes in cardiac structure and function in heart failure with preserved ejection fraction and the impact of spironolactone. *Circ Heart Fail*. 2015;8(6):1052-1058. doi:[10.1161/CIRCHEARTFAILURE.115.002249](https://doi.org/10.1161/CIRCHEARTFAILURE.115.002249)

11. Mehra MR, Uriel N, Naka Y, et al; MOMENTUM 3 Investigators. A fully magnetically levitated left ventricular assist device—final report. *N Engl J Med*. 2019;380(17):1618-1627. doi:[10.1056/NEJMoa1900486](https://doi.org/10.1056/NEJMoa1900486)

12. Rogers JG, Pagani FD, Tatoes AJ, et al. Intrapericardial left ventricular assist device for advanced heart failure. *N Engl J Med*. 2017;376(5):451-460. doi:10.1056/NEJMoa1602954
13. Sitbon O, Bosch J, Cottreel E, et al. Macitentan for the treatment of portopulmonary hypertension (PORTICO): a multicenter, randomized, double-blind, placebo-controlled, phase 4 trial. *Lancet Respir Med*. 2019;7(7):594-604. doi:10.1016/S2213-2600(19)30091-8
14. Galiè N, Jansa P, Pulido T, et al. SERAPHIN hemodynamic substudy: the effect of the dual endothelin receptor antagonist macitentan on hemodynamic parameters and NT-proBNP levels and their association with disease progression in patients with pulmonary arterial hypertension. *Eur Heart J*. 2017;38(15):1147-1155. doi:10.1093/eurheartj/ehx025
15. Ghofrani HA, Galiè N, Grimminger F, et al; PATENT-1 Study Group. Riociguat for the treatment of pulmonary arterial hypertension. *N Engl J Med*. 2013;369(4):330-340. doi:10.1056/NEJMoa1209655
16. Chin KM, Sitbon O, Doelberg M, et al. Three- vs 2-drug therapy for patients with newly diagnosed pulmonary arterial hypertension. *J Am Coll Cardiol*. 2021;78(14):1393-1403. doi:10.1016/j.jacc.2021.07.057
17. Humbert M, McLaughlin VV, Badesch DB, et al; ZENITH Trial Investigators. Sotatercept in patients with pulmonary arterial hypertension at high risk for death. *N Engl J Med*. 2025;392(20):1987-2000. doi:10.1056/NEJMoa2415160
18. Goudie AR, Lipworth BJ, Hopkinson PJ, Wei L, Struthers AD. Tadalafil in patients with chronic obstructive pulmonary disease: a randomized, double-blind, parallel-group, placebo-controlled trial. *Lancet Respir Med*. 2014;2(4):293-300. doi:10.1016/S2213-2600(14)70013-X
19. Ghofrani HA, D'Armini AM, Grimminger F, et al; CHEST-1 Study Group. Riociguat for the treatment of chronic thromboembolic pulmonary hypertension. *N Engl J Med*. 2013;369(4):319-329. doi:10.1056/NEJMoa1209657
20. Ghofrani HA, Simonneau G, D'Armini AM, et al; MERIT study investigators. Macitentan for the treatment of inoperable chronic thromboembolic pulmonary hypertension (MERIT-1): results from the multicenter, phase 2, randomized, double-blind, placebo-controlled study. *Lancet Respir Med*. 2024;12(4):e21-e30. doi:10.1016/S2213-2600(24)00027-4
21. Tapson VF, Sterling K, Jones N, et al. A randomized trial of the optimum duration of acoustic pulse thrombolysis procedure in acute intermediate-risk pulmonary embolism: the OPTALYSE PE trial. *JACC Cardiovasc Interv*. 2018;11(14):1401-1410. doi:10.1016/j.jcin.2018.04.008
22. Sadeghipour P, Jenab Y, Moosavi J, et al. Catheter-directed thrombolysis vs anticoagulation in patients with acute intermediate-high-risk pulmonary embolism: the CANARY randomized clinical trial. *JAMA Cardiol*. 2022;3591. doi:10.1001/jamacardio.2022.3591
23. Kucher N, Boekstegers P, Müller OJ, et al. Randomized, controlled trial of ultrasound-assisted catheter-directed thrombolysis for acute intermediate-risk pulmonary embolism. *Circulation*. 2014;129(4):479-486. doi:10.1161/CIRCULATIONAHA.113.005544
24. Avgerinos ED, Jaber W, Lacomis J, et al; SUNSET sPE Collaborators. Randomized trial comparing standard versus ultrasound-assisted thrombolysis for submassive pulmonary embolism: the SUNSET SPE trial. *JACC Cardiovasc Interv*. 2021;14(12):1364-1373. doi:10.1016/j.jcin.2021.04.049
25. Barrios D, Durán D, Rodríguez C, et al; Air vs Oxygen for Intermediate-Risk Pulmonary Embolism Investigators. Oxygen therapy in patients with intermediate-risk acute pulmonary embolism: a randomized trial. *Chest*. 2024;165(3):673-681. doi:10.1016/j.chest.2023.09.007
26. Jaber WA, Gonsalves CF, Storteky S, et al; PEERLESS Committees and Investigators. Large-bore mechanical thrombectomy vs catheter-directed thrombolysis in the management of intermediate-risk pulmonary embolism: primary results of the PEERLESS randomized controlled trial. *Circulation*. 2025;151(5):260-273. doi:10.1161/CIRCULATIONAHA.124.072364
27. Pastore MC, Stefanini A, Mandoli GE, et al. Dapagliflozin effects on cardiac deformation in heart failure and secondary clinical outcome. *JACC Cardiovasc Imaging*. 2024;17(12):1399-1408. doi:10.1016/j.jcmg.2024.05.014
28. Il'Giovine ZJ, Mulder H, Chiswell K, et al. Right ventricular longitudinal strain reproducibility using vendor-dependent and vendor-independent software. *J Am Soc Echocardiogr*. 2018;31(6):721-732.e5. doi:10.1016/j.echo.2018.01.008
29. Mirea O, Berceanu M, Donoiu I, Militaru C, Săftoiu A, Istrătoae O. Variability of right ventricular global and segmental longitudinal strain measurements. *Echocardiography*. 2019;36(1):102-109. doi:10.1111/echo.14218
30. Genovese D, Mor-Avi V, Palermo C, et al. Comparison between 4-chamber and right ventricular-focused views for the quantitative evaluation of right ventricular size and function. *J Am Soc Echocardiogr*. 2019;32(4):484-494. doi:10.1016/j.echo.2018.11.014
31. Pieske B, Maggioni AP, Lam CSP, et al. Vericiguat in patients with worsening chronic heart failure and preserved ejection fraction: results of the Soluble Guanylate Cyclase Stimulator in Heart Failure Patients with PRESERVED EF (SOCRATES-PRESERVED) study. *Eur Heart J*. 2017;38(15):1119-1127. doi:10.1093/eurheartj/ehw593
32. Solomon SD, Claggett B, Lewis EF, et al; TOPCAT Investigators. Influence of ejection fraction on outcomes and efficacy of spironolactone in patients with heart failure with preserved ejection fraction. *Eur Heart J*. 2016;37(5):455-462. doi:10.1093/eurheartj/ehv464
33. Solomon SD, Ostrominski JW, Wang X, et al; STEP-HFpEF Trial Committees and Investigators. Effect of semaglutide on cardiac structure and function in patients with obesity-related heart failure. *J Am Coll Cardiol*. 2024;84(17):1587-1602. doi:10.1016/j.jacc.2024.08.021
34. Reddy YNV, Carter RE, Sorimachi H, et al. Dapagliflozin and right ventricular-pulmonary vascular interaction in heart failure with preserved ejection fraction: a secondary analysis of a randomized clinical trial. *JAMA Cardiol*. 2024;9(9):843-851. doi:10.1001/jamacardio.2024.1914
35. Lampert BC, Teuteberg JJ. Right ventricular failure after left ventricular assist devices. *J Heart Lung Transplant*. 2015;34(9):1123-1130. doi:10.1016/j.healun.2015.06.015
36. Desai MY, Wolski K, Owens A, et al; VALOR-HCM Investigators. Mavacamten in patients with hypertrophic cardiomyopathy referred for septal reduction: week 128 results from VALOR-HCM. *Circulation*. 2025;151(19):1378-1390. doi:10.1161/CIRCULATIONAHA.124.072445
37. Hegde SM, Claggett BL, Wang X, et al; SEQUOIA-HCM Investigators. Impact of aficamten on echocardiographic cardiac structure and function in symptomatic obstructive hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 2024;84(19):1789-1802. doi:10.1016/j.jacc.2024.08.002
38. Jering KS, Fontana M, Lairez O, et al. Effects of vutrisiran on cardiac structure and function in patients with transthyretin amyloidosis with cardiomyopathy: secondary outcomes of the HELIOS-B trial. *Nat Med*. 2025;31(10):3560-3568. doi:10.1038/s41591-025-03851-z
39. Jering KS, Fontana M, Skali H, et al. Effects of vutrisiran on cardiac function and outcomes in patients with transthyretin amyloidosis with cardiomyopathy. *J Am Coll Cardiol*. 2025;86(6):444-455. doi:10.1016/j.jacc.2025.06.022
40. Doldi PM, Stolz L, Kalbacher D, et al; EuroSMR and PRIME-MR Investigators. Right ventricular dysfunction predicts outcome after transcatheter mitral valve repair for primary mitral valve regurgitation. *Eur J Heart Fail*. 2022;24(11):2162-2171. doi:10.1002/ehfj.2661
41. Karam N, Stolz L, Orban M, et al. Impact of right ventricular dysfunction on outcomes after transcatheter edge-to-edge repair for secondary mitral regurgitation. *JACC Cardiovasc Imaging*. 2021;14(4):768-778. doi:10.1016/j.jcmg.2020.12.015
42. Sorajja P, Whisenant B, Hamid N, et al; TRILUMINATE Pivotal Investigators. Transcatheter repair for patients with tricuspid regurgitation. *N Engl J Med*. 2023;388(20):1833-1842. doi:10.1056/NEJMoa2300525
43. von Bardeleben RS, Lurz P, Sorajja P, et al; TRILUMINATE Trial Investigators. Two-year outcomes for tricuspid repair with a transcatheter edge-to-edge valve repair from the transatlantic TRILUMINATE trial. *Circ Cardiovasc Interv*. 2023;16(8):e012888. doi:10.1161/CIRCINTERVENTIONS.122.012888
44. Stolz L, Weckbach LT, Glaser H, et al. Biphasic right ventricular reverse remodeling following tricuspid valve transcatheter edge-to-edge repair. *JACC Cardiovasc Interv*. 2024;17(23):2811-2822. doi:10.1016/j.jcin.2024.09.069
45. Hahn RT, Makkar R, Thourani VH, et al; TRISCEND II Trial Investigators. Transcatheter valve replacement in severe tricuspid regurgitation. *N Engl J Med*. 2025;392(2):115-126. doi:10.1056/NEJMoa2401918
46. Xu H, Chen M, Wang Z, et al. Midterm outcomes of K-Clip transcatheter tricuspid annuloplasty system in patients with severe functional tricuspid regurgitation. *JACC Cardiovasc Interv*. 2024;17(23):2796-2807. doi:10.1016/j.jcin.2024.10.007
47. Zhang X, Jin Q, Li W, et al. Transcatheter annuloplasty with the K-Clip system for tricuspid regurgitation: 1-year results from the Tristar study. *EuroIntervention*. 2025;21(5):e262-e271. doi:10.4244/EIJ-D-24-00591

48. Pan X, Lu F, Wang Y, et al. Transcatheter tricuspid valve replacement with the novel system: 1-year outcomes from the TRAVEL study. *JACC Cardiovasc Interv*. 2025;18(10):1276-1285. doi:10.1016/j.jcin.2024.12.030
49. Lurz P, Rommel KP, Schmitz T, et al. Real-world 1-year results of tricuspid edge-to-edge repair from the BRIGHT Study. *J Am Coll Cardiol*. 2024;84(7):607-616. doi:10.1016/j.jacc.2024.05.006
50. Stolz L, Kresoja KP, von Stein J, et al; on behalf of the EuroTR Investigators. Residual tricuspid regurgitation after tricuspid transcatheter edge-to-edge repair: insights into the EUROTR registry. *Eur J Heart Fail*. 2024;26(8):1850-1860. doi:10.1002/ehf.3274
51. Howard LS. Prognostic factors in pulmonary arterial hypertension: assessing the course of the disease. *Eur Respir Rev*. 2011;20(122):236-242. doi:10.1183/09059180.00006711
52. Shah AM, Campbell P, Rocha GQ, et al; IMPRES Investigators. Effect of imatinib as add-on therapy on echocardiographic measures of right ventricular function in patients with significant pulmonary arterial hypertension. *Eur Heart J*. 2015;36(10):623-632. doi:10.1093/eurheartj/ehu035
53. Souza R, Badesch DB, Ghofrani HA, et al. Effects of sotatercept on hemodynamics and right heart function: analysis of the STELLAR trial. *Eur Respir J*. 2023;62(3):2301107. doi:10.1183/13993003.01107-2023
54. Waxman AB, Systrom DM, Manimaran S, de Oliveira Pena J, Lu J, Rischard FP. SPECTRA phase 2b study: impact of sotatercept on exercise tolerance and right ventricular function in pulmonary arterial hypertension. *Circ Heart Fail*. 2024;17(5):e011227. doi:10.1161/CIRCHEARTFAILURE.123.011227
55. Vonk Noordegraaf A, Channick R, Cottreel E, et al. The REPAIR study: effects of macitentan on RV structure and function in pulmonary arterial hypertension. *JACC Cardiovasc Imaging*. 2022;15(2):240-253. doi:10.1016/j.jcmg.2021.07.027
56. Rahaghi FF, Humbert M, Hoepfer MM, et al. Future treatment paradigms in pulmonary arterial hypertension: a personal view from physicians, health authorities, and patients. *Lancet Respir Med*. 2025;13(4):364-370. doi:10.1016/S2213-2600(24)00425-9
57. Konstantinides SV, Meyer G, Becattini C, et al; ESC Scientific Document Group. 2019 ESC guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). *Eur Heart J*. 2020;41(4):543-603. doi:10.1093/eurheartj/ehz405
58. Hanmandlu A, Mannari A, Abidov A. Diagnostic and prognostic value of RV strain defined as RV/LV ratio on chest CT in acute pulmonary embolism: a systematic review. *Echocardiography*. 2025;42(3):e70107. doi:10.1111/echo.70107
59. Mavromanolis AC, Barco S, Agno W, et al; PEITHO-2 Investigators. Recovery of right ventricular function after intermediate-risk pulmonary embolism: results from the multicenter Pulmonary Embolism International Trial (PEITHO)-2. *Clin Res Cardiol*. 2023;112(10):1372-1381. doi:10.1007/s00392-022-02138-4
60. Yuriditsky E, Zhang RS, Ahuja T, Bangalore S, Horowitz JM. The latest in the management of pulmonary embolism. *Breathe (Sheff)*. 2025;21(2):240100. doi:10.1183/20734735.0100-2024
61. Surkova E, Brida M, Muraru D, et al. Three-dimensional echocardiography in adults with congenital heart disease: a scientific statement of the European Society of Cardiology Working Group on Adult Congenital Heart Disease and the European Association of Cardiovascular Imaging of the European Society of Cardiology. *Eur Heart J Cardiovasc Imaging*. 2025;26(7):1244-1269. doi:10.1093/ehjci/jeaf105
62. Bokma JP, Winter MM, van Dijk AP, et al. Effect of losartan on right ventricular dysfunction: results from the double-blind, randomized REDEFINE trial (Right Ventricular Dysfunction in Tetralogy of Fallot: Inhibition of the Renin-Angiotensin-Aldosterone System) in adults with repaired tetralogy of Fallot. *Circulation*. 2018;137(14):1463-1471. doi:10.1161/CIRCULATIONAHA.117.031438
63. Babu-Narayan SV, Uebing A, Davlouros PA, et al. Randomized trial of ramipril in repaired tetralogy of Fallot and pulmonary regurgitation: the APPROPRIATE study (Ace Inhibitors for Potential Prevention of the Deleterious Effects of Pulmonary Regurgitation in Adults With Repaired Tetralogy of Fallot). *Int J Cardiol*. 2012;154(3):299-305. doi:10.1016/j.ijcard.2010.09.057
64. Geva T, Gauvreau K, Powell AJ, et al. Randomized trial of pulmonary valve replacement with and without right ventricular remodeling surgery. *Circulation*. 2010;122(11)(suppl):S201-S208. doi:10.1161/CIRCULATIONAHA.110.951178
65. Greutmann M, Tobler D, Engel R, et al; SERVE Investigators. Effect of phosphodiesterase-5 inhibition on systemic right ventricular size and function: a multicenter, double-blind, randomized, placebo-controlled trial: SERVE. *Eur J Heart Fail*. 2023;25(7):1105-1114. doi:10.1002/ehf.2924
66. van der Bom T, Winter MM, Bouma BJ, et al. Effect of valsartan on systemic right ventricular function: a double-blind, randomized, placebo-controlled pilot trial. *Circulation*. 2013;127(3):322-330. doi:10.1161/CIRCULATIONAHA.112.135392
67. Fusco F, Scognamiglio G, Abbate M, et al. Dapagliflozin in patients with a failing systemic right ventricle: results from the DAPA-SERVE trial. *JACC Heart Fail*. 2024;12(4):789-791. doi:10.1016/j.jchf.2024.01.006
68. Benz DC, Gräni C, Antiochos P, et al. Cardiac magnetic resonance biomarkers as surrogate end points in cardiovascular trials for myocardial diseases. *Eur Heart J*. 2023;44(45):4738-4747. doi:10.1093/eurheartj/ehad510
69. Kovacs A, Magunia H, Nicoara A, et al. Challenges and opportunities in assessing right ventricular structure and function: a roadmap for standardization, clinical implementation and research. *Nat Rev Cardiol*. 2026;23(1):60-77. doi:10.1038/s41569-025-01180-9
70. US Food and Drug Administration. Table of surrogate end points that were the basis of drug approval or licensure. Accessed October 22, 2025. <https://www.fda.gov/drugs/development-resources/table-surrogate-endpoints-were-basis-drug-approval-or-licensure>
71. Tolvaj M, Kovács A, Radu N, et al. Significant disagreement between conventional parameters and 3D echocardiography-derived ejection fraction in the detection of right ventricular systolic dysfunction and its association with outcomes. *J Am Soc Echocardiogr*. 2024;37(7):677-686. doi:10.1016/j.echo.2024.04.005
72. Mukherjee M, Rudski LG, Addetta K, et al. Guidelines for the echocardiographic assessment of the right heart in adults and special considerations in pulmonary hypertension: recommendations from the American Society of Echocardiography. *J Am Soc Echocardiogr*. 2025;38(3):141-186. doi:10.1016/j.echo.2025.01.006
73. Holste G, Oikonomou EK, Tokodi M, Kovács A, Wang Z, Khera R. Complete AI-enabled echocardiography interpretation with multitask deep learning. *JAMA*. 2025;334(4):306-318. doi:10.1001/jama.2025.8731
74. Szijártó Á, Merkely B, Kovács A, Tokodi M; QUEST-EF Investigators. Deep learning-enabled echocardiographic assessment of biventricular ejection fractions: the dual-task QUEST-EF model. *Eur Heart J Cardiovasc Imaging*. 2025;26(8):1402-1405. doi:10.1093/ehjci/jeaf147